Northern University of Business and Technology, Khulna

Department of Computer Science and Engineering

Frontend Progress Report

Week 07 Assignment

AI Code Security Analyzer

(Pluto AI)

Course Title	Software Development Project
Course Code	CSE4104
Section	7C
Team	CSE4104-7C-T04
Project Title	AI Code Security Analyzer (Pluto AI)
Assignment	Week 07 – Frontend Development and Back-end Integration

Submitted To

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1. Team Members

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2. Project Overview

AI Code Security Analyzer (Pluto AI) is a web-based application that performs AI-assisted source code security analysis. The frontend has been developed using React.js and integrated with the backend APIs developed during the previous phase. The application provides user authentication, project management, security scanning, dashboard visualization, and vulnerability reporting through an intuitive interface.

3. Objectives

The primary objectives of the frontend development phase are:

- Develop a responsive user interface using React.js.
- Integrate the frontend with backend REST APIs.
- Implement JWT-based user authentication.
- Provide an intuitive dashboard for project management.
- Enable AI-powered source code vulnerability analysis.
- Improve overall usability through responsive design and validation.

4. Frontend Technology Stack

Technology	Purpose
React.js	Frontend framework
Vite	Build tool
Tailwind CSS	UI Styling
Axios	Backend API communication
React Router	Navigation
JWT Authentication	Secure user authentication
Node.js + Express	Backend APIs
PostgreSQL	Database

5. Project Structure

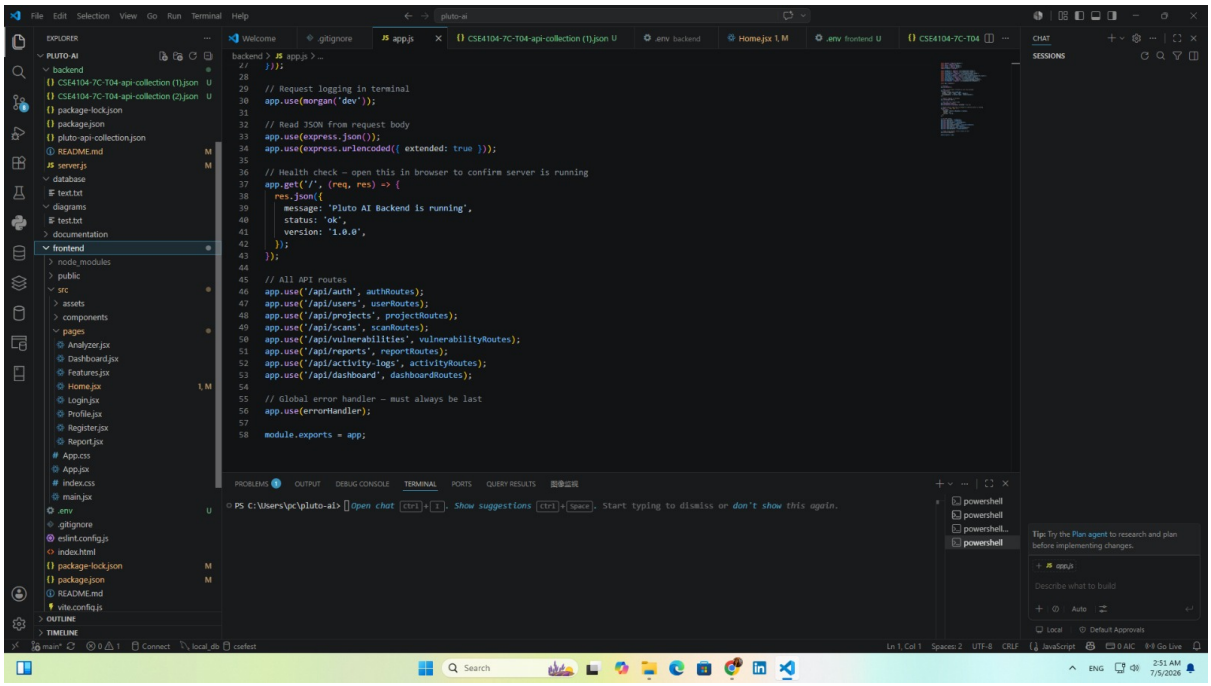


Figure 1: Project directory structure of the AI Code Security Analyzer.

6. Form Validation

Client-side validation has been implemented for all user forms to improve data quality and user experience.

The validation features include:

- Required field validation
- Email format validation
- Password length verification
- Password confirmation matching
- Invalid credential messages
- Successful operation notifications

7. User Experience Improvements

Several user experience enhancements have been implemented to improve usability.

- Responsive navigation bar
- Loading indicators
- Success notifications
- Error messages
- Dynamic dashboard statistics
- Activity history
- Simple and intuitive navigation

8. Completed Frontend Pages

Landing Page	Completed
Features Page	Completed
Login	Completed
Registration	Completed
Dashboard	Completed
AI Security Scanner	Completed
Profile	Completed
Responsive UI	Completed

9. Backend API Integration

The frontend communicates dynamically with the backend developed during Week 06. RESTful APIs were integrated using Axios to exchange JSON data between the client and server. The implemented API integrations include:

- User Registration
- User Login
- JWT Authentication
- Dashboard Statistics
- Project Management
- Security Scan Submission
- Activity Logs
- AI Vulnerability Analysis

All user information and scan results displayed in the interface are retrieved dynamically from the backend rather than using static data.

10. Authentication Flow

JWT-based authentication was implemented to secure the application. After successful registration or login, the backend generates a JSON Web Token (JWT), which is stored on the client side and attached to subsequent API requests. Protected routes such as the Dashboard and Security Scanner can only be accessed by authenticated users.

1. User submits login credentials.
2. Backend validates the credentials.
3. JWT token is generated.
4. Token is stored in browser local storage.
5. Protected pages become accessible.
6. User logout removes the stored token.

11. Responsive Design

The user interface was tested on desktop and mobile layouts. Responsive navigation, adaptive cards, flexible forms, and scalable layouts ensure usability across different screen sizes.

12. Development Progress

The frontend implementation has been completed and integrated successfully with backend APIs. Authentication, dashboard visualization, project management, and AI-powered code analysis functionalities are fully operational.

13. GitHub Repository

`https://github.com/0xSaikat/pluto-ai`

14. User Interface Screenshots

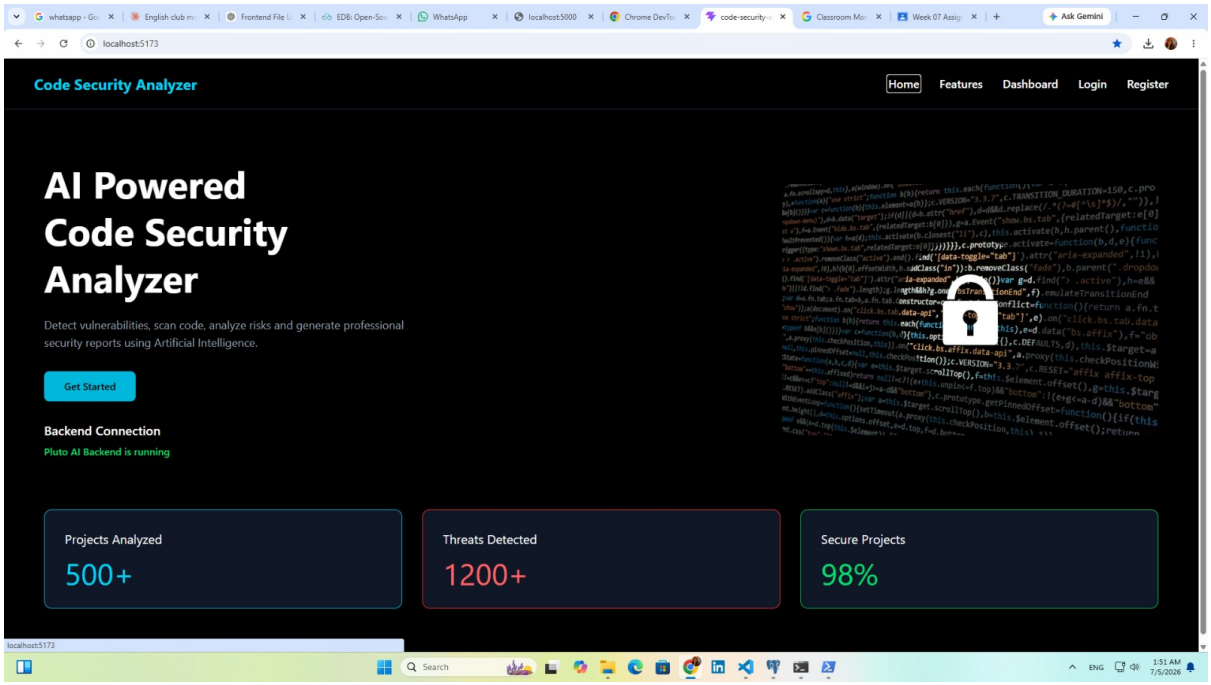


Figure 2: Landing (Home) page of the AI Code Security Analyzer.

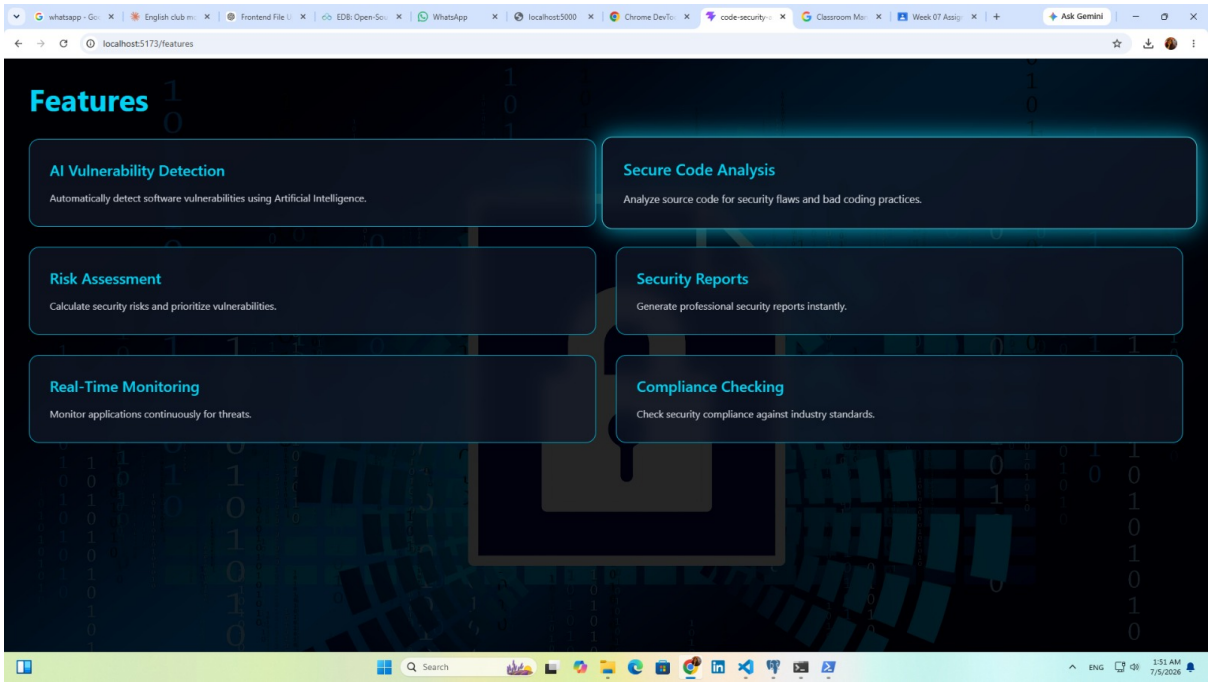


Figure 3: System Features page highlighting the core functionalities of the application.

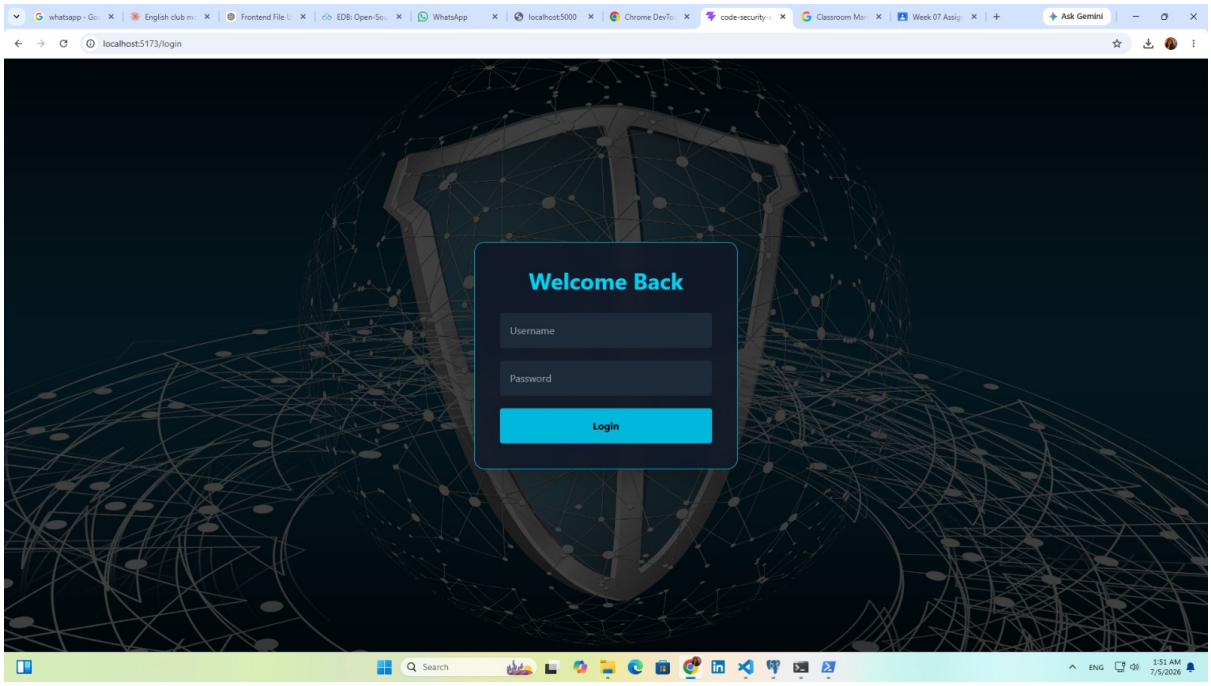


Figure 4: User Login page providing secure authentication.

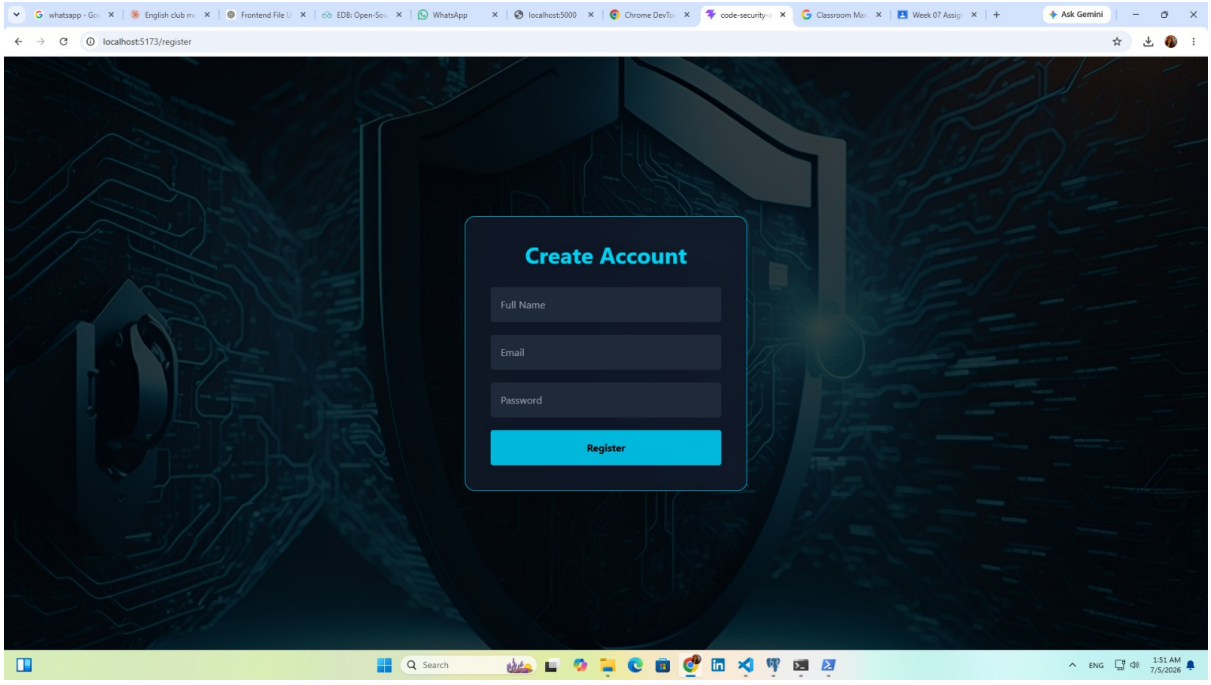


Figure 5: User Registration page integrated with backend authentication APIs.

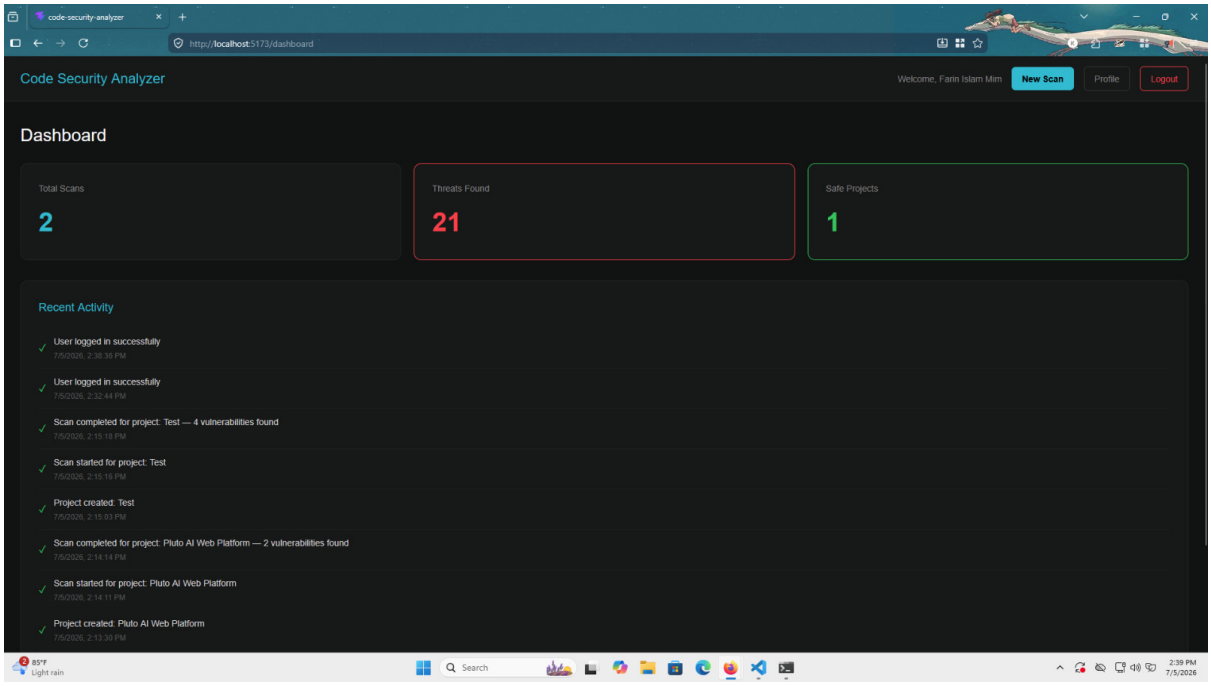


Figure 6: User Dashboard displaying scan statistics, threats, secure projects, and activity logs.

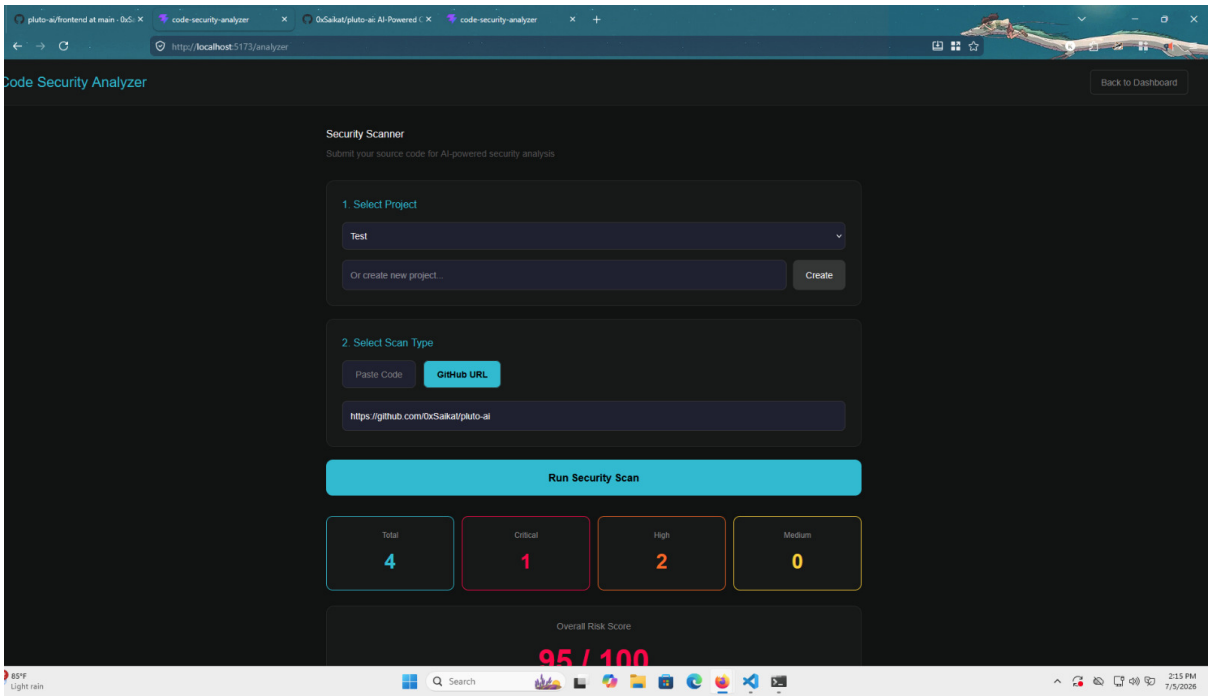


Figure 7: AI Security Scanner interface supporting GitHub repository and source code analysis.

15. Conclusion

The frontend of the AI Code Security Analyzer has been successfully implemented according to the approved UI/UX design and integrated with the backend services. The application supports secure authentication, AI-powered code scanning, dashboard analytics, responsive design, and a user-friendly interface. The modular architecture ensures maintainability and future scalability.